

XT 电子纸相框通讯协议

XT ePaper Frame

Communication Protocol

V1.2

Version 版本号	Modified By 修改人	Date 修改日期	Description 说明
1.0	Lobos	20250717	开放基础版
1.1	Lobos	20250910	加入附件一、设备修改服务器通讯参数
1.2	Lobos	20251118	修正部分描述
1.3	Song	20260410	新增协议

1. 说明

- 数据交换方式：硬件设备与服务器通信采用MQTT协议，基于发布订阅的模式进行数据交换，默认使用小写字符。
- 上行数据：指硬件设备报给服务器的数据。
- 下行数据：指服务端下发给硬件设备的数据。

2. 通讯接口说明 Communication Interface Description

2.1 服务器通讯：Server Communication

采用 MQTT 协议 v3.1.1 版本，QOS：1。

MQTT Topic:

上行数据发送 使用 Topic: /device/report/{clientid}

下行数据接收 使用 Topic: /inkjoyap/{clientid}

Uses MQTT protocol version 3.1.1, QOS: 1.

。 MQTT Topics:

Uplink data transmission uses Topic: /device/report/{clientid}

Downlink data reception uses Topic: /inkjoyap/{clientid}

2.2 设备与App蓝牙通信消息定义

概述：定义了 手机 App 与 ESP32 设备 在蓝牙连接阶段（目前只有配网阶段）的 [Blufi](#) 自定义消息格式。

- 。 所有消息均以 JSON 格式 进行定义，字符采用 UTF-8 编码。
- 。 Blufi 数据帧(分片)最大支持 64 K 的数据内容
- 。 Blufi 相关通信代码实现均以乐鑫官方 Blufi demo 为基线来对齐
 - 。 Blufi 示例: [bluetooth/blufi](#)：（需烧录至 ESP32）
 - 。 手机应用程序: EspBlufi
 - 。 Android 版本: [EspBlufi For Android](#)
 - 。 IOS 版本: [EspBlufi For iOS](#)

Blufi 协议详见官方文档

<https://docs.espressif.com/projects/espressif-esp32-api-guides/ble/blufi.html>

2.2.1 消息方向

方向	说明
App → Device	手机端发送指令到设备
Device → App	设备回复执行结果或状态

2.2.2 数据结构通用字段

字段名	类型	方向	说明
action	string	双向	消息动作类型
msgid	string	App → Device	消息ID, 毫秒级时间戳
data	object	双向	具体数据内容 (结构由 action 决定)
data.result	int	Device → App	指令执行结果(如果有): 0=成功, 非0=错误码
data.ack_msgid	string	Device → App	应答的消息ID

2.2.3 设置 MQTT 配置信息

请求 (App → Device)

```
{
  "msgid": "1756108400173", // 消息ID, 使用毫秒级时间戳
  "action": "mqtt_config", // 消息动作
  "data": { // 消息体
    "host": "mqtt.example.com",
    "port": 9000,
    "usr": "user123",
    "pwd": "p@ssw0rd"
  }
}
```

回复 (Device → App)

```
{
  "action": "mqtt_config_ack", // 消息动作
  "data": { // 消息体
    "result": 0, // 指令执行结果: 0 = 成功, 非0 = 具体错误码
    "ack_msgid": "1756108400173", // 应答的消息ID
    // 可添加其他需要回传的数据
  }
}
```

2.2.4 设置WiFi国家码 (Country Code)

用于根据地区差异调整 Wi-Fi 信道范围、发射功率。

乐鑫相关文档:

[Wi-Fi 国家/地区代码](#)

[esp_wifi_set_country_code](#)

Supported country codes are "01"(world safe mode) "AT","AU","BE","BG","BR","CA","CH","CN","CY","CZ","DE","DK","EE","ES","FI","FR","GB","GR","HK","HR","HU","IE","IN","IS","IT","JP","KR","LI","LT","LU","LV","MT","MX","NL","NO","NZ","PL","PT","RO","SE","SI","SK","TW","US"

The third octet of country code string is one of the following: ' ', 'O', 'I', 'X', otherwise it is considered as ' '.

请求 (App → Device)

```
{
  "msgid": "1756108400180",
  "action": "wifi_set_country_code",
  "data": {
    "country": "JP" // ISO 3166-1 alpha-2 国家码, 例如: CN, US, JP
  }
}
```

回复 (Device → App)

```
{
  "action": "wifi_set_country_code_ack",
  "data": {
    "result": 0, // 指令执行结果: 0 = 成功, 非0 = 具体错误码
    "pending": 1, // 如需暂缓执行, 回传 pending 字段。App 在解析时, 会认为它的
    优先级高于 result
    "ack_msgid": "1756108400180"
    // 可添加其他需要回传的数据
  }
}
```

3. 消息

2.1 登录 Login

APP扫码设备配网成功后，上报一次数据

上行数据: Uplink Data:

Key		Type	Values	Description
action		string	login	Message action
clientid		string		Device ID
msgid		string		Message ID, uses millisecond timestamp
stamac		string		MAC Address
data	ver	string		Firmware Version
	statype	string		Base Station Type 1、 LINUX Base Station 2、 Wi-Fi Base Station 3、 Integrated Base Station & Tag 4、 PC Base Station (USB)

```
JSON
{
  "action": "login",
  "clientid": "A61F5AE3A12D"
  "msgid": "1712800547780",
  "stamac": "A61F5AE3A12D",
  "data": {
    "ver": "xtap_001",
    "statype": "1"
  }
}
```

下行数据: Downlink Data:

Key		Type	Values	Description
action		string	login_ack	Message action
msgid		string		Message ID, uses millisecond timestamp
stamac		string		MAC Address
date	ack_msgid	string		Message ID of the response

```
JSON
{
  "action": "login_ack",
  "msgid": "1750172946729",
  "stamac": "6055F9FCE598"
  "data": {
    "ack_msgid": "1750172946729"
  }
}
```

2.2.心跳上报 Heartbeat

上行数据: Uplink Data:

Key		Type	Values	Description
action		string	heart	Message actions
clientid		string		Device ID
msgid		string		Message ID, Uses millisecond timestamp
stamac		string		MAC Address
data	statype	int	type	Base Station Type 1、 LINUX Base Station 2、 Wi-Fi Base Station 3、 Integrated Base Station & Tag 4、 PC Base Station (USB)
	ack	int		Response flag: (0: No Ack, 1: Ack)
	wifi_ch	int		
	wifi	string		WiFi switch on/off
	ble	string		Bluetooth switch on/off
	tf	string		TF card status: present/absent
	tfsize	decimal		TF card capacity, in MB
	tfused	decimal		Used space on the TF card, in MB
	orientation	int		方向字段 当前取不到预留! 0: 默认方向 (Portrait, 竖屏) 90: 向左横屏 (Landscape, 设备顶部在左边) 180: 倒立 (Upside-down) 270: 向右横屏 (Landscape, 设备顶部在右边)
	battery	int		Battery percentage (0-100, as an integer)
	wifi_rssi	int		WiFi signal strength (in dBm, negative integers)
	wifi_name	string		Wi-Fi name
	ble_rssi	int		Bluetooth signal strength (in dBm, negative integers)
	version	string		ESP firmware version
	country_code	string		Current country code
	fpga_ver	string		FPGA version

JSON

```
{
  "action": "heart",
  "clientid": "30EDA0E44624",
  "msgid": "72074003686",
  "stamac": "30EDA0E44624",
  "data": {
    "type": 3,
    "ack": 1,
    "wifi_ch": 1,
    "wifi": "on",
    "ble": "off",
    "tf": "absent",
    "tfsize": 0,
    "tfused": 0,
    "orientation": 0,
    "battery": 86,
    "wifi_rssi": -46,
    "wifi_name": "aa",
    "ble_rssi": 0,
    "version": "0.1.2",
    "country_code": "1",
    "fpga_ver": "0.0.1"
  }
}
```

下行数据: Downlink Data:

服务器收到上报数据后自动下发一条消息,告知接收到

Key`		Type	Values	Description
action		string	heart_ack	Message actions
msgid		string		Message ID, Uses millisecond timestamp
stamac		string		MAC Address
date	ack_msgid	string		

```
JSON
{
  "action": "heart_ack",
  "msgid": "1750172946729",
  "stamac": "6055F9FCE598",
  "data": {
    "ack_msgid": "1750172946729"
  }
}
```

2.3 OTA指令 OTA instructions

下行数据: Downlink Data:

Key		Type	Values	Description
action		string	ota	Message actions
msgid		string		Message ID, Uses millisecond timestamp
stamac		string		MAC Address
data	host	string		Server Domain Name or IP Address
	port	int		Port
	path	string		File path and filename

```
{
  "action": "ota",
  "msgid": "7911d3429cc846",
  "stamac": "A61F5AE3A12D",
  "data": {
    "host": "10.41.1.84",
    "port": 25801,
    "path": "/ota/20240411/3ca9db5ad"
  }
}
```

上行数据: Uplink Data:

通知服务端OTA执行结果

Key		Type	Values	Description
action		string	ota_ack	Message actions
clientid		string		Device ID
msgid		string		Message ID, Uses millisecond timestamp
stamac		string		MAC Address
data	result	int		执行结果 参考文档末尾的状态码列表
	ack_msgid	string		Message ID of the response

```
{
  "action": "ota_ack",
  "clientid": "A61F5AE3A12D",
  "msgid": "1712800547780",
  "stamac": "A61F5AE3A12D",
  "data": {
    "result": 1,
    "ack_msgid": "1712800547780"
  }
}
```

2.4 FPGA指令 FPGA instructions

下行数据: Uplink Data:

Key		Type	Values	Description
action		string	fpga	Message actions
msgid		string		Message ID, Uses millisecond timestamp
stamac		string		MAC Address
data	host	string		Server Domain Name or IP Address
	port	int		Port
	path	string		File path and filename

```
{
  "action": "fpga",
  "msgid": "7911d3429cc846",
  "stamac": "A61F5AE3A12D",
  "data": {
    "host": "10.41.1.84",
    "port": 25801,
    "path": "/fpga/20240411/3ca9db5ad"
  }
}
```

上行数据: Uplink Data:

通知服务端FPGA执行结果

```
{
  "action": "fpga_ack",
  "clientid": "A61F5AE3A12D",
  "msgid": "1712800547780",
  "stamac": "A61F5AE3A12D",
  "data": {
    "result": 1,
    "ack_msgid": "1712800547780"
  }
}
```

2.5 Wifi系列模式配置指令 Configuration commands for Wi-Fi series modes

下行数据: Uplink Data:

Key		Type	Values	Description
action		string	wifimode	Message actions
msgid		string		Message ID, Uses millisecond timestamp

stamac		string		MAC Address
data	mode	int		功能码 0x01实时通讯模式 0x02定时通讯模式 0x03休眠模式
	live	int		定时唤醒后保持时间(秒) 取值范围：60-65535 默认值：600
	timingtime	int		定时通讯参数：定时时间(分钟) 取值范围：60-1440 默认值：240

```
{
  "action": "wifimode",
  "msgid": "1712800547780",
  "stamac": "A61F5AE3A12D",
  "data": {
    "mode": 1,
    "live": 600,
    "timingTime": 240
  }
}
```

上行数据：Uplink Data:

通知服务端Wifi系列模式配置结果

Key	Type	Values	Description
action	string	wifimode_ack	Message actions
clientid	string		Device ID
msgid	string		Message ID, Uses millisecond timestamp
stamac	string		MAC Address
data	result	int	执行结果 参考文档末尾的状态码列表
	ack_msgid	string	应答的消息ID
	mode	int	功能码 0x01实时通讯模式 0x02定时通讯模式 0x03停机模式

```
{  
  "action": "wifimode_ack",  
  "clientid": "A61F5AE3A12D",  
  "msgid": "1712800547780",  
  "stamac": "A61F5AE3A12D",  
  "data": {  
    "result": 1,  
    "ack_msgid": "1712800547780",  
    "mode": 1  
  }  
}
```

2.6 Wifi定时休眠配置指令 Commands for configuring Wi-Fi sleep scheduling

下行数据: Uplink Data:

Key		Type	Values	Description
action		string	wifi_sleep	Message actions
msgid		string		Message ID, Uses millisecond timestamp
stamac		string		MAC Address
data	mode	int		休眠模式 0:关闭休眠 1:一次性 2:周期
	live	int		休眠开始时间 APP保存的是当地时间，下发时会统一转为UTC时区的时间 mode=1时格式:yyyy-MM-dd HH:mm mode=2时格式:HH:mm
	timingtime	int		休眠截止时间 APP保存的是当地时间，下发时会统一转为UTC时区的时间 mode=1时格式:yyyy-MM-dd HH:mm mode=2时格式:HH:mm

```
{
  "action": "wifimode",
  "msgid": "1712800547780",
  "stamac": "A61F5AE3A12D",
  "data": {
    "mode": 1,
    "beginTime": "",
    "endTime": ""
  }
}
```

上行数据: Uplink Data:

通知服务端Wifi休眠配置结果

Key		Type	Values	Description
action		string	wifi_sleep_ack	Message actions
clientid		string		Device ID
msgid		string		Message ID, Uses millisecond timestamp
stamac		string		MAC Address
data	result	int		执行结果 参考文档末尾的状态码列表
	ack_msgid	string		应答的消息ID

```
{
  "action": "wifi_sleep_ack",
  "clientid": "A61F5AE3A12D",
  "msgid": "1712800547780",
  "stamac": "A61F5AE3A12D",
  "data": {
    "result": 1,
    "ack_msgid": "1712800547780"
  }
}
```

2.7 设备配置下发指令 Issue device configuration commands

下行数据: Uplink Data:

Key		Type	Values	Description
action		string	device_config	Message actions
msgid		string		Message ID, Uses millisecond timestamp
stamac		string		MAC Address
data	batchkey	int		A random 12-character string
	deviceId	int		The first 12 digits of the device ID
	uid	int		User ID

```
{
  "action": "device_config",
  "msgid": "1712800547780",
  "stamac": "A61F5AE3A12D",
  "data": {
    "batchKey": "",
    "deviceId": "",
    "uid": ""
  }
}
```

上行数据: Uplink Data:

通知服务端设备配置下发指令

Key		Type	Values	Description
action		string	device_config_ack	Message actions
clientid		string		Device ID
msgid		string		Message ID, Uses millisecond timestamp
stamac		string		MAC Address
data	result	int		执行结果 参考文档末尾的状态码列表
	ack_msgid	string		应答的消息ID

```
{
  "action": "device_config_ack",
  "clientid": "A61F5AE3A12D",
  "msgid": "1712800547780",
  "stamac": "A61F5AE3A12D",
  "data": {
    "result": 1,
    "ack_msgid": "1712800547780"
  }
}
```

2.8 播放单图指令 Playlist image instructions

目前服务端定时触发单图下发

下行数据: Uplink Data:

Key			Type	Values	Description
action			string	play	Message actions
msgid			string		Message ID, Uses millisecond timestamp
stamac			string		MAC Address
data	host		string		Server domain name or IP address
	port		int		port
	imgs	imgid	string		Document ID
		imgurl	string		File path and filename

```
{
  "action": "play",
  "msgid": "1750169235376",
  "stamac": "3A:84:43:F6:F4:85",
  "data": {
    "host": "101.201.79.63",
    "port": 9000,
    "imgs": [{
      "imgid": "a6d134e2-e71e-44e6-be99-eb34f1c2af37",
      "imgurl": "/ufile/hh7man.bin"
    }]
  }
}
```

上行数据: Uplink Data:

通知服务端播放指令的执行情况

Key		Type	Values	Description
action		string	play_ack	Message actions
clientid		string		Device ID
msgid		string		Message ID, Uses millisecond timestamp
stamac		string		MAC Address
data	result	int		执行结果 参考文档末尾的状态码列表
	ack_msgid	string		应答的消息ID

```
{
  "action": "play_ack",
  "clientid": "A61F5AE3A12D",
  "msgid": "1712800547780",
  "stamac": "A61F5AE3A12D",
  "data": {
    "result": 1,
    "ack_msgid": "1712800547780"
  }
}
```

2.9 播放策略指令 Playback policy instructions

服务端下发更新时间,定时到服务端拉图

下行数据: Uplink Data:

Key		Type	Values	Description
action		string	strategy	Message actions
msgid		string		Message ID, Uses millisecond timestamp
stamac		string		MAC Address
data	idle	int		轮播后是否休眠 0:休眠 1:不休眠
	strategy	int		播放策略 1: 顺序播放 2: 随机播放
	host	string		服务器域名或IP地址
	port	int		端口
	path	string		拉图路径
	updatetype	int		更新时间类型 1: 固定时间 2: 间隔时间
	begintime	string		间隔时间-开始时间
	endtime	string		间隔时间-截止时间
	intervalminutes	int		间隔时间-间隔分钟数
	updatedays	int		固定时间-更新频率(天)
	updatetimest	string array		固定时间-更新时分

```
{
  "action": "strategy",
  "msgid": "1750169235376",
  "stamac": "3A:84:43:F6:F4:85",
  "data": {
    "idle": 1,
    "strategy": 1,
    "host": "101.201.79.63",
    "port": 9000,
    "path": "/a/a.bin",
    "updatetype": 1,
    "begintime": "18:00",
    "endtime": "19:00",
    "intervalminutes": 60,
    "updatedays": 2,
    "updatetimest": ["11:00", "15:00"]
  }
}
```

上行数据: Uplink Data:

通知服务端播放策略指令的执行情况，可以应答多个，每次更换图片时可以上报一条消息

Key		Type	Values	Description
action		string	strategy_ack	Message actions
clientid		string		Device ID
msgid		string		Message ID, Uses millisecond timestamp
stamac		string		MAC Address
data	result	int		执行结果 参考文档末尾的状态码列表
	ack_msgid	string		应答的消息ID

```
{
  "action": "strategy_ack",
  "clientid": "A61F5AE3A12D",
  "msgid": "1712800547780",
  "stamac": "A61F5AE3A12D",
  "data": {
    "result": 1,
    "ack_msgid": "1712800547780"
  }
}
```


2.10 播放策略+Bin指令 Playback policy + Bin command

服务端下发更新时间和所有图片bin的列表,设备定时在本地根据策略计算要显示哪张图

下行数据: Uplink Data:

Key		Type	Values	Description
action		string	strategy	Message actions
msgid		string		Message ID, Uses millisecond timestamp
stamac		string		MAC Address
data	idle	int		轮播后是否休眠 0:休眠 1:不休眠
	strategy	int		播放策略 1: 顺序播放 2: 随机播放
	host	string		服务器域名或IP地址
	port	int		端口
	path	string		查询所有图片地址的路径
	updatetype	string		更新时间类型 1: 固定时间 2: 间隔时间
	begintime	string		间隔时间-开始时间
	endtime	string		间隔时间-截止时间
	intervalminutes	int		间隔时间-间隔分钟数
	updatedays	int		固定时间-更新频率(天)
	updatetimelist	string array		固定时间-更新时分

// 消息示例

```
{
  "action": "strategy_bin",
  "msgid": "1750169235376",
  "stamac": "3A:84:43:F6:F4:85",
  "data": {
    "idle": 1,
    "strategy": 1,
    "host": "101.201.79.63",
    "port": 9000,
    "path": "/b/xxxx",
    "updatetype": 1,
    "begintime": "18:00",
    "endtime": "19:00",
    "intervalminutes": 60,
    "updatedays": 2,
    "updatetimelist": ["11:00", "15:00"]
  }
}
```

// 查询所有图片地址路径返回的数据格式

```
{
  "code": 0,
  "msg": "success",
  "data": {
    "host": "101.201.79.63",
    "port": 8080,
    "imgList": [
      "/inkjoy/a/b5b698ec-630e-4627-84dc-a372681d9215",
      "/inkjoy/a/f262e914-d080-4dfa-b0e4-b52c764681bc",
      "/inkjoy/a/ec408172-8f28-4667-8462-0c47d8761709",
      "/inkjoy/a/ef086b68-2b9b-4e22-a18b-52ca0f75824f"
    ]
  }
}
```

2.11 暂停轮播 Pause carousel

服务端删除当前生效的播放策略（对应播放策略、播放策略+Bin 这两种）后，下发此消息通知给设备

下行数据: Uplink Data:

Key	Type	Values	Description
action	string	strategy_stop	Message actions
msgid	string		Message ID, Uses millisecond timestamp
stamac	string		MAC Address

```
// 消息示例
{
  "action": "strategy_stop",
  "msgid": "1750169235376",
  "stamac": "3A:84:43:F6:F4:85"
}
```

上行数据: Uplink Data:

通知服务端是否停止成功

Key		Type	Values	Description
action		string	strategy_stop_ack	Message actions
clientid		string		Device ID
msgid		string		Message ID, Uses millisecond timestamp
stamac		string		MAC Address
data	result	int		执行结果 参考文档末尾的状态码列表
	ack_msgid	string		应答的消息ID

```
{
  "action": "strategy_stop_ack",
  "clientid": "A61F5AE3A12D",
  "msgid": "1712800547780",
  "stamac": "A61F5AE3A12D",
  "data": {
    "result": 1,
    "ack_msgid": "1712800547780"
  }
}
```

2.12 播放TF卡内图片 Play images from the TF card

通知设备播放TF卡内图片

下行数据: Uplink Data:

Key	Type	Values	Description
action	string	play_tf	Message actions
msgid	string		Message ID, Uses millisecond timestamp
stamac	string		MAC Address

```
// 消息示例
{
  "action": "play_tf",
  "msgid": "1750169235376",
  "stamac": "3A:84:43:F6:F4:85"
}
```

上行数据: Uplink Data:

通知服务端是否成功执行

Key		Type	Values	Description
action		string	play_tf_ack	Message actions
clientid		string		Device ID
msgid		string		Message ID, Uses millisecond timestamp
stamac		string		MAC Address
data	result	int		执行结果 参考文档末尾的状态码列表
	ack_msgid	string		应答的消息ID

```
{
  "action": "play_tf_ack",
  "clientid": "A61F5AE3A12D",
  "msgid": "1712800547780",
  "stamac": "A61F5AE3A12D",
  "data": {
    "result": 1,
    "ack_msgid": "1712800547780"
  }
}
```

2.13 清空TF卡内容 Erase the contents of the TF card

通知设备清空TF卡内容

下行数据: Uplink Data:

Key		Type	Values	Description
action		string	clean_tf	Message actions
msgid		string		Message ID, Uses millisecond timestamp
stamac		string		MAC Address
date	type	string		清空范围 ALL=全部,LOCAL=内置,OTHER=其他

```
// 消息示例
{
  "action": "clean_tf",
  "msgid": "1750169235376",
  "stamac": "3A:84:43:F6:F4:85",
  "data": {
    "type": "ALL"
  }
}
```

上行数据: Uplink Data:

通知服务端是否成功执行

Key		Type	Values	Description
action		string	clean_tf_ack	Message actions
clientid		string		Device ID
msgid		string		Message ID, Uses millisecond timestamp
stamac		string		MAC Address
data	result	int		执行结果 参考文档末尾的状态码列表
	ack_msgid	string		应答的消息ID

```
{
  "action": "clean_tf_ack",
  "clientid": "A61F5AE3A12D",
  "msgid": "1712800547780",
  "stamac": "A61F5AE3A12D",
  "data": {
    "result": 1,
    "ack_msgid": "1712800547780"
  }
}
```

2.14 下载内置图片 Download the built-in images

通知设备下载内置图片

下行数据: Uplink Data:

Key		Type	Values	Description
action		string	down_int_img	Message actions
msgid		string		Message ID, Uses millisecond timestamp
stamac		string		MAC Address
data	host	string		Server domain name or IP address
	port	int		port
	path	string		Retrieve the paths of all image URLs

```
// 消息示例
```

```
{
  "action": "down_int_img",
  "msgid": "1750169235376",
  "stamac": "3A:84:43:F6:F4:85",
  "data": {
    "host": "101.201.79.63",
    "port": 8080,
    "path": "/b/xxxx",
  }
}
```

```
// 查询所有图片地址路径返回的数据格式
```

```
/*
```

```
FACTORY=出厂页面
```

```
SET=设置页面
```

```
NO_WIFI=无wifi提示
```

```
LOW_BATTERY=低电量提示
```

```
LOW_POWER_OFF=低电量关机页面
```

```
CONNECTED=连接页面
```

数据格式:[x,y,w,h,sum,path]//[坐标X,坐标Y,图片宽,图片高,图片数据CRC16校验值,图片地址]如坐标X、坐标Y都为0,则表示使用默认坐标

```
*/
```

```
{
  "code": 0,
  "msg": "success",
  "data": {
    "host": "mf.ink.00r.top",
    "port": 443,
    "bins": {
      "SET": [
        "0",
        "0",
        "1200",
        "1600",

```

```
        "21455",
        "/offical/config/intimg/img_set.bin"
    ],
    "FACTORY": [
        "0",
        "0",
        "632",
        "432",
        "207",
        "/offical/config/intimg/img_factory.bin"
    ],
    "NO_WIFI": [
        "0",
        "0",
        "912",
        "1168",
        "46240",
        "/offical/config/intimg/img_no_wifi.bin"
    ],
    "CONNECTED": [
        "0",
        "0",
        "1200",
        "1600",
        "xxx",
        "/offical/config/intimg/img_connected.bin"
    ],
    "LOW_BATTERY": [
        "0",
        "0",
        "632",
        "106",
        "46331",
        "/offical/config/intimg/img_low_batt.bin"
    ],
    "LOW_POWER_OFF": [
        "0",
        "0",
        "632",
        "106",
        "50686",
        "/offical/config/intimg/img_low_power_off.bin"
    ]
}
}
```

上行数据: Uplink Data:

通知服务端是否成功执行

Key		Type	Values	Description
action		string	down_int_img_ack	Message actions
clientid		string		Device ID
msgid		string		Message ID, Uses millisecond timestamp
stamac		string		MAC Address
data	result	int		执行结果 参考文档末尾的状态码列表
	ack_msgid	string		应答的消息ID

```
{
  "action": "down_int_img_ack",
  "clientid": "A61F5AE3A12D",
  "msgid": "1712800547780",
  "stamac": "A61F5AE3A12D",
  "data": {
    "result": 1,
    "ack_msgid": "1712800547780"
  }
}
```

2.15 刷新图片 Refresh image

通知设备清空TF卡内容

下行数据: Uplink Data:

Key	Type	Values	Description
action	string	image_refresh	Message actions
msgid	string		Message ID, Uses millisecond timestamp
stamac	string		MAC Address

```
// 消息示例
{
  "action": "image_refresh",
  "msgid": "1750169235376",
  "stamac": "3A:84:43:F6:F4:85"
}
```

上行数据: Uplink Data:

Key	Type	Values	Description
action	string	image_refresh_ack	Message actions
msgid	string		Message ID, Uses millisecond timestamp
stamac	string		MAC Address

```
// 消息示例
{
  "action": "image_refresh_ack",
  "msgid": "1750169235376",
  "stamac": "3A:84:43:F6:F4:85"
}
```

2.16 国家码变更通知 Notice of Change to the National Code

通知设备国家码变更

下行数据: Uplink Data:

Key		Type	Values	Description
action		string	country	Message actions
msgid		string		Message ID, Uses millisecond timestamp
stamac		string		MAC Address
date	country_code	string		National Code

```
// 消息示例
{
  "action": "country",
  "msgid": "1750169235376",
  "stamac": "3A:84:43:F6:F4:85",
  "data": {
    "country_code": "1"
  }
}
```

上行数据: Uplink Data:

Key		Type	Values	Description
action		string	country_ack	Message actions
clientid		string		Device ID
msgid		string		Message ID, Uses millisecond timestamp
stamac		string		MAC Address
data	result	int		执行结果 参考文档末尾的状态码列表
	ack_msgid	string		应答的消息ID

```
{
  "action": "country_ack",
  "clientid": "A61F5AE3A12D",
  "msgid": "1712800547780",
  "stamac": "A61F5AE3A12D",
  "data": {
    "result": 1,
    "ack_msgid": "1712800547780"
  }
}
```

2.17 关机通知 Shutdown notification

通知服务器设备即将关机

上行数据: Uplink Data:

Key	Type	Values	Description
action	string	shutdown	Message actions
msgid	string		Message ID, Uses millisecond timestamp
stamac	string		MAC Address

```
// 消息示例
{
  "action": "shutdown",
  "msgid": "1750169235376",
  "stamac": "3A:84:43:F6:F4:85"
}
```

下行数据: Uplink Data:

通知设备收到

Key		Type	Values	Description
action		string	shutdown_ack	Message actions
clientid		string		Device ID
msgid		string		Message ID, Uses millisecond timestamp
stamac		string		MAC Address
date	ack_msgid	int		应答的消息ID

```
{
  "action": "shutdown_ack",
  "clientid": "A61F5AE3A12D",
  "msgid": "1712800547780",
  "stamac": "A61F5AE3A12D",
  "data": {
    "ack_msgid": "1712800547780"
  }
}
```

2.18 设备标签上报 Device station tag reporting

上行数据: Uplink Data:

Key	Type	Values	Description
action	string	login	Message actions
clientid	string		Device ID
msgid	string		Message ID, Uses millisecond timestamp
stamac	string		MAC Address
data	int array		蓝牙标签信号强度: 1字节 标签MAC地址: 6字节 其他上报信息: 参考下图

上报类型 (1B)	硬件版本 (1B)	软件版本 (2B)	设备编码 (2B)	电量 (1B)	发射功率 (1B)	安全等级 (1B)	参数类型 (1B)	参数 (≥14B)
0-255	0-255			0-100				

2.上报类型:标签上报类型(兼容旧版本以区分广播包数据结构;本新协议:0x03:漫游上报 0x04:

标签上报(带唤醒)0xFE:可连接广播0xFD:标签上报(无需唤醒)0xFC:重连广播;其它:保留)

3.硬件版本:产品的硬件版本,高 4位表示主版本号(0-F), 低4位表示次版本号(0-F)

4.软件版本: 左高右低,高字节高 4位主版本号(0-F),低4位次版本号(0-F),低字节位(0-255)。

例如:0x1002, 表示版本 1.0.2

5.设备编码:设备的编码DD, 作为查询产品信息的标识

6.电量:设备电池电量百分比, 0~100%

7.发射功率:高 4bit 表示芯片型号, 低 4bit 表示发射功率

8.安全等级: 预留目0x01:不认证不加密 0x02 :license认证不加密 0x03:认证+加密

9.参数类型&参数:

类型	参数
0x00	默认不带参数
0x01	分组, 2Bytes参数, 表示分组号
0x02	MAC地址, 6Bytes 参数

```
{
  "action": "uptag",
  "clientid": "A61F5AE3A12D",
  "msgid": "1712799763896",
  "stamac": "A61F5AE3A12D",
  "data": [170, 49, 56, 5, 0, 111, 52, 253, 1, 16, 16, 0, 3, 87, 6, 255, 255]
}
```

4. 状态码定义表 Status Code Definition Table

Description	Code
收到下行数据. 需要 ack. 并且无异常情况 Downlink received, ACK required, no error	100
获取 FLASH 地址失败 Failed to get FLASH address	101
获取 FTP 服务器失败 Failed to get FTP server	102
获取文件大小失败 Failed to get file size	103
下载文件失败 Failed to download file	104
添加设备列表失败 Failed to add device to list	105
指令添加成功 Command added successfully	106
下载图片或文件成功 Image/file downloaded successfully	107
连接蓝牙设备超时 Bluetooth device connection timeout	110
指令执行失败 Command execution failed	112
执行成功 Execution successful	113
文件发送失败 File transmission failed	114
STA 获取 flash 地址失败 STA failed to get flash address	115
STA 获取文件大小失败 STA failed to get file size	116
STA 添加设备列表失败 STA failed to add device to list	117
STA 文件发送失败 STA file transmission failed	118
下载成功 Download successful	255
文件发送到蓝牙主机失败 File transmission to Bluetooth host failed	109

5. 附件一. 设备修改服务器通讯参数

Appendix I. Device Modifies Server Communication Parameters

方法 1:

设备进入配网模式(操作见使用说明书), 设备会建立一个 WIFI 热点, 连接其热点后, 通过浏览器访问设置页面可设置设备参数。

Method 1

Enter the device's network configuration mode (see user manual).

The device will create a Wi-Fi hotspot. Connect to this hotspot and access the configuration page via a web browser to set device parameters.

操作过程如下:

通过电脑或手机连接设备热点, 热点名称为"XT_ESP_" + (设备 MAC 地址 后 4 位), 密码为 1234567890 连接后, 浏览器访问 192.168.4.1,

服务器主要修改这 4 项: Server IP、Server Port、User、Password, 修改后点 SET 进行保存。如要立刻重启可在 Sys Reset 栏中点击 RESET 重启设备。

Procedure:

1. Connect your computer or phone to the device's hotspot.

Hotspot Name: "XT_ESP_" + (last 4 digits of device MAC address)

Password: 1234567890

2. After connecting, open a browser and visit 192.168.4.1.

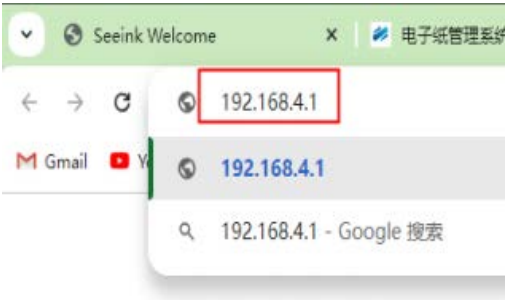
3. Modify the following 4 main server parameters: Server IP, Server Port, User, Password. Click SET to save.

4. To reboot immediately, click RESET in the Sys Reset section.

热点名称



浏览器地址



服务器配置部分参数

Server Set

Server Type:

MQTT

Server IP:

101.201.79.63

Server Port:

1883

User:

device

Password:

KeepALIVE s:

600

DTIM 100ms:

10

ClientID:

30EDA0E3FBB4

Cert1:

...

Cert2:

...

SET

方法 2

设备进入配网模式(操作见使用说明书)，手机可连接 BLE 设备进行设置，BLE 设备名 IJ_(MAC 地址)，可通过 BluFi 协议的自定义协议，发送配置数据 来修改服务器通讯参数。

Method 2

Enter the device's network configuration mode (see user manual). A mobile phone can connect to the BLE device for configuration.

BLE Device Name: IJ_(MAC Address)

Server communication parameters can be modified by sending configuration data via the custom protocol within the BluFi protocol.

注意：设备版本 V0.1.1 及以上才支持

注意：设备版本 V0.2.7 及以上"broker"字段已修改为"host" 主要修改这 4 个参数："host" "port" "usr" "pwd",

Note: Supported from device firmware version V0.1.1 and above.

Note: The field "broker" has been changed to "host" from firmware version V0.2.7 and above.

The main parameters to modify are: "host", "port", "usr", "pwd".

参考数据：Reference Data:

```
JSON
{
  "msgid": "1756108400173",
  "action": "mqtt_config",
  "data": {
    "host": "192.168.1.1",
    "port": 1883,
    "usr": "device",
    "pwd": "1234567890ab"
  }
}
```

使用 EspBluFi 示例操作过程如下：

The operation process using the EspBluFi example app is as follows:

